

Technology Stack

Date	1 october2026
Team ID	SWTID-2026-6275
Project Name	Legal Ease: AI-Powered Legal Document Generator
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	React.js with HTML5 / CSS3 (Tailwind CSS)	Component-based UI gives reusable forms, a live document preview and a responsive layout that works on desktop and mobile.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python with FastAPI	Fast asynchronous processing, easy integration with AI/NLP libraries, and Python is familiar to the team.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	PostgreSQL	Strong relational mapping for users, document templates and generated documents, with reliable and secure storage.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	AWS (EC2, S3) with Docker	Easy scalability and cost-effective hosting; Docker gives consistent deployment and S3 stores generated documents.
5	Version Control & CI/CD	Git and GitHub with GitHub Actions	Seamless team collaboration through branches and pull requests, with automated testing and deployment.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	LLM API (e.g., OpenAI GPT) and python-docx / ReportLab	Generates legal clauses and drafts from user inputs, and exports the final document as DOCX or PDF.